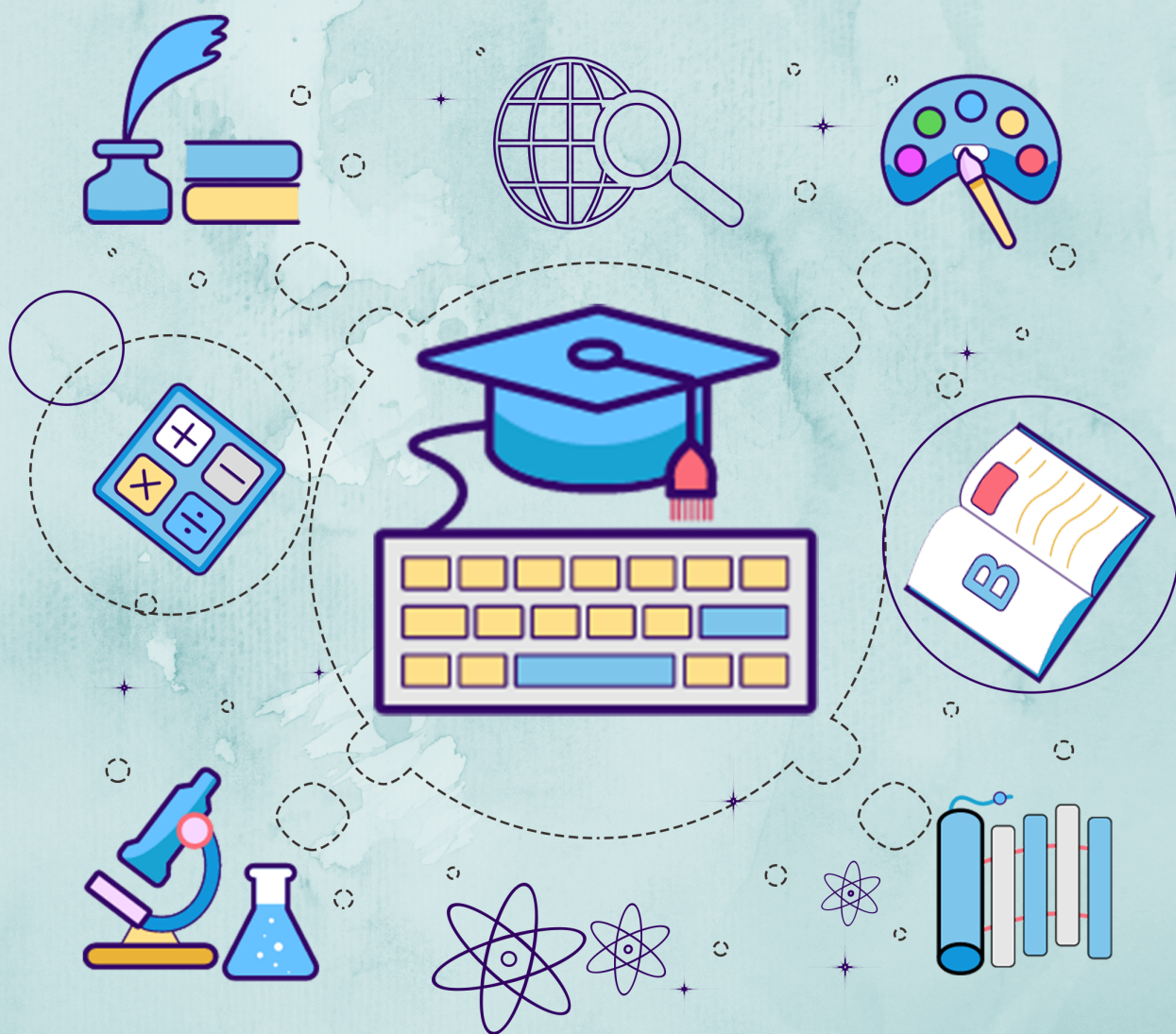


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SUPERCONDUCTIVITY & PHOTONICS

A. Superconductivity

The phenomenon of loss of resistance of certain materials when cooled below certain specific temperature is called superconductivity and the materials showing this property are called superconductors.

The phenomenon of superconductivity was discovered by Kamerlingh Onnes in 1911. Eg: For mercury ($T_c = 4.2\text{K}$) ; for aluminium ($T_c = 1.175\text{K}$)

Properties of superconductors

(i) Effect of temperature: Transition temperature (T_c)

The temperature at which a normal material turns into a superconductor is called critical temperature or transition temperature..

(ii) Effect of magnetic field: Critical magnetic field (H_c)

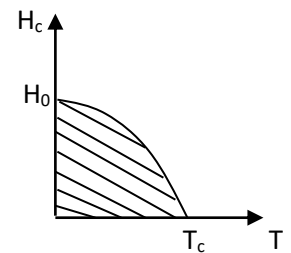
The maximum magnetic field that can be applied across the material without destroying its superconducting nature is called critical field H_c .

The material acts as a superconductor, when temperature is less than transition temperature and field is less than critical field. When $H < H_c$ and $T < T_c$, metals acts as a superconductor

- Variation of critical field H_c and temperature is shown below.

$$H_c(T) = H_c(0) \left[1 - \frac{T^2}{T_c^2} \right] \text{ where } H_c(T) - \text{critical field at } T\text{K};$$

$$H_c(0) - \text{critical field at } 0\text{K}; \quad T_c - \text{transition temperature}$$



(iii) Effect of current density: Critical current I_c (Silsbee effect)

- The maximum current that can be passed through a superconducting material without destroying the superconducting property is called critical current I_c . $I_c = 2\pi r H_c$

where H_c is critical magnetic field and r is radius of superconducting rod (wire)

- The maximum current that can be passed through a superconductor per unit area of cross section,

without destroying the superconducting property is called critical current density J_c . $J_c = \frac{I_c}{A} = \frac{2\pi r H_c}{\pi r^2} = \frac{2H_c}{r}$

(iv) Isotope effect:

- Discovered by Maxwell and Reynold in 1950.
- The variation of transition temperature with isotopic mass is called isotope effect.
- Transition temperature and isotopic mass are inversely proportional. $T_c \propto \frac{1}{M^\alpha}$

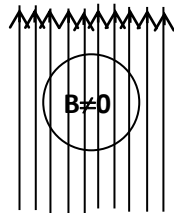
where M is isotopic mass and $\alpha = \frac{1}{2}$ (a constant)

$$T_c \times M^{1/2} = \text{a constant}$$

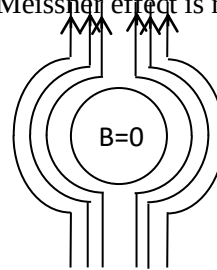
$$T_c \sqrt{M} = \text{a constant}$$

Meissner Effect / Perfect Diamagnetism

It was discovered by Meissner and Ochenfeld in 1933. The phenomenon of expulsion of magnetic field lines from the interior of a super conductor when it is cooled below the transition temperature in the presence of magnetic field, is known as Meissner effect. so the field inside the superconductor is zero. Meissner effect is reversible.



$T > T_c$ and $H > H_c$
Normal conductor



$T < T_c$ and $H < H_c$
Super conductor

We have Flux density, $B = \mu_0 (M + H)$ where μ_0 is permeability of free space; M – magnetization

From Meissner effect, $B = 0$ inside the super conductor at $T < T_c$

Then $\mu_0 (M + H) = 0$ or $M = -H$

Magnetic susceptibility $\chi = \frac{M}{H} = -1$

For diamagnets, magnetic susceptibility $\chi = -1$

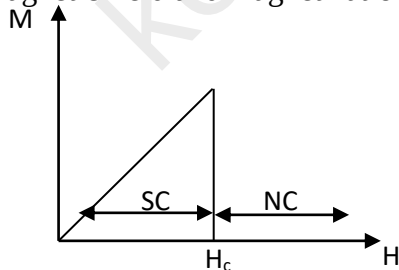
Thus super conductor is a perfect diamagnet.

Types of superconductor

Depending upon on the magnetic properties in an applied magnetic field, superconductors are divided into two types

Type I superconductors: (Soft superconductor)

* Variation of magnetic field and magnetization for type I superconductor is shown below.



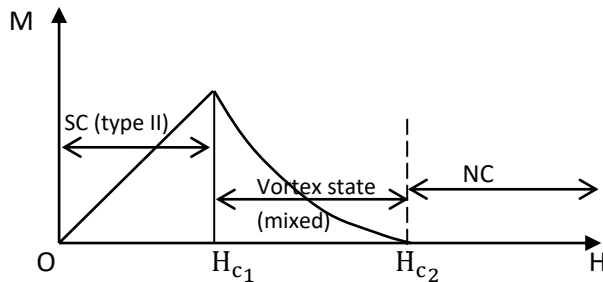
- * Type I superconductors strictly follow Meissner effect
- * When field H increases, magnetization M also increases linearly up to H_c .
- * Below critical field H_c , material is a super conductor and above H_c material is a normal conductor.
- * Transition from superconducting state to normal state is a sudden process.
- * H_c is very small. i.e., $H_c \approx 0.1\text{T}$ to 0.2T .

So a small magnetic field is only required to destroy the superconducting nature of the material. So Type I superconductors are known as soft superconductors.

- * At the field H_c transition is reversible.
- * Eg: Lead, Tin, Aluminium, Mercury, Magnesium etc

Type II superconductors: (Hard superconductor)

- * Variation of magnetic field and magnetization for type II superconductor is shown below.



- * When external field H increases, magnetization M also increases linearly up to H_{c1} (lower critical field).
- * Type II superconductors do not follow Meissner effect strictly.
- * There are two critical fields, lower and upper fields
- * Below H_{c1} material is a super conductor Beyond H_{c2} , the material changes to normal conductor.
- * The state in between H_{c1} and H_{c2} is called vortex or intermediate state. (mixed state).
- * The transition from superconducting state to normal conducting state is very slow and gradual.
- * The value of H_{c2} is high . i.e., $H_{c2} \approx 100T$

So high magnetic field is required to destroy superconducting nature. So type II superconductor is called **hard** superconductor.

- * Transition is irreversible .
- * Eg: Niobium, Vanadium, Silicon, Niobium - tin, Niobium – titanium, Germanium

Uses

- * Type II superconductors are used in power generators.
- * Because type II superconductors can carry very high current densities, they have great technological importance.
- * The high magnetic fields produced by type II superconductors are used in particle accelerators, plasma production, fusion reaction etc.
- * This strong magnetic field is used for magnetic levitation.

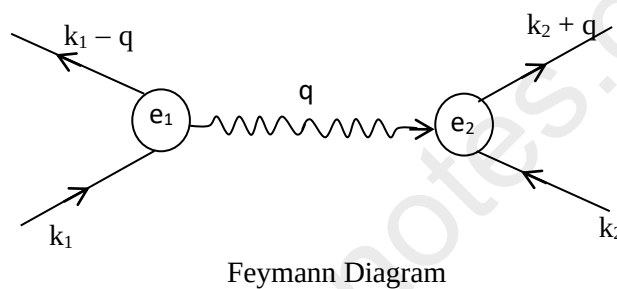
BCS theory

In 1957, Bardeen, Cooper and Schriffer developed a new theory to explain superconductivity called BCS theory. It is based on the formation of Cooper pair of electrons.

According to BCS theory superconductivity is due to the attractive interaction between electrons at very low temperature. Due to this interaction electrons move in pairs called cooper pairs. The two electrons in cooper pair move coherently through lattice without any collision with lattice ions so that resistance becomes zero and material becomes superconductor.

When an electron e_1 moving through the lattice it distorts the lattice. Because of coulomb's interaction, the surrounding ions will be attracted towards that electron. Due to displacement of ions, the positive charge density increases around that electron e_1 . If another electron e_2 happens to pass closely to e_1 then e_1 will be shielded from e_2 due to condensation of positive charge density around it and e_2 will be attracted in that direction. The two electrons appear as if they are bonded to each other and behave like a single particle called cooper pair. The two electrons in a Cooper pair are exchanging phonons through lattice ions and shown in figure.

Here, an electron $\bar{e}_1$, with a wave vector k_1 , emits a phonon and change its state to $(k_1 - q)$. A second electron $\bar{e}_2$ with a wave vector k_2 absorbs that a phonon and change its state to $(k_2 + q)$.



Characteristics of cooper Pairs

- * Two electrons in a Cooper pair have opposite momenta and opposite spin.
- * The mass of a Cooper pair is $2m$ where m is the effective mass of the electron. The charge on Cooper pair is $-2e$.
- * As the spin of Cooper pair is zero, the Cooper pair behaves like a Boson, and it does not obey Pauli's exclusion principle.
- * At a temperature less than T_c , almost all free electrons are paired as Cooper pair. Above T_c , this pairing is broken.
- * The binding energy of Cooper pair is of the order of $10^{-3} - 10^{-4} \text{ eV}$. It is slightly less than twice the energy of free electron.

Coherence length (ϵ)

The maximum length up to which the states of paired electrons are correlated to produce superconductivity is called coherence length. $\epsilon_0 = \frac{h v_f}{4\pi\Delta}$ where v_f fermi velocity and 2Δ is energy gap.

Josephson Effect

Josephson suggested that a super current consisting of Cooper pairs can be made to flow across an insulating gap between two superconductors provided the gap is small enough. Such a junction is called

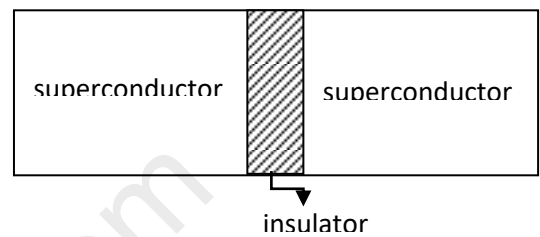
Josephson junction or weak link. This tunneling of Cooper pairs across the Josephson junction is called Josephson effect.

DC Josephson Effect:

Cooper pairs can tunnel from one superconductor to the other separated by a thin layer of insulator. As a result of this tunneling, there is a net dc current flows across the junction even in the absence of any external electric or magnetic field. This is called DC Josephson effect.

$$\text{Current across the junction is } I_s = I_{\max} \sin \phi_0$$

where $I_{\max}$ - maximum current at zero voltage.



AC Josephson Effect:

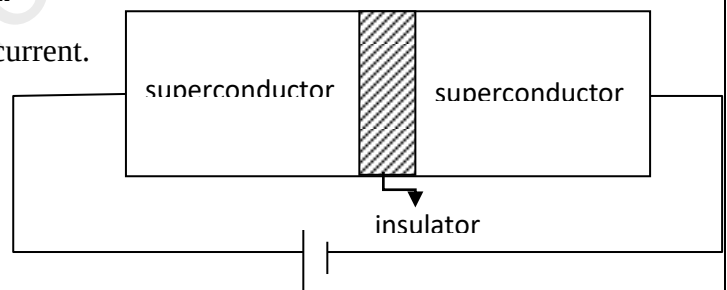
When a dc voltage is applied across the Josephson junction, the junction generates an alternating current at a frequency of $\frac{2eV}{h}$. This is called AC Josephson effect.

where $2e$ - charge of a Cooper pair and V is the applied dc voltage.

Josephson current,

$$I_s = I_{\max} \sin(\phi_0 + \delta) \quad . \quad \text{Here } \delta = \omega t = \frac{2\pi \times 2eV}{h} t = \frac{4\pi eVt}{h}$$

Hence $I_s = I_{\max} \sin\left(\phi_0 + \frac{4\pi eVt}{h}\right) t$. This is ac current.



Applications of Superconductors

(i) Superconductors are used to produce a very strong and powerful magnetic field in the order of 20T. This high magnetic field is used in particle accelerators, cyclotrons, controlled nuclear fusion etc.

(ii) Medical application:

- * They are used in MRI.
- * Superconducting magnetic field is used to remove tumour cell from the healthy cells.
- * Group of squids are used for the diagnosis of epilepsy.

(iii) Electronic and small devices:

- * Squid
- * Frictionless bearing, magnetically controlled superconducting switches, superconductor fuses, breakers, superconducting transformers.

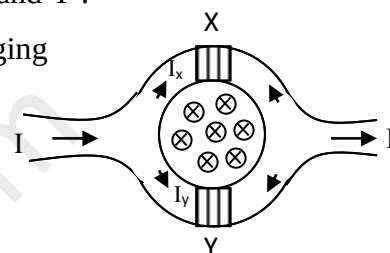
(iv) Computers:

- * High capacity and high speed computer chips can be developed with superconductors.
- * Used to perform logic and store functions in computers.

(v) Low loss transmission lines can be made with superconductors.

SQUID (Superconducting Quantum Interference Device)

- * Squid is a very sensitive instrument used for measuring small changes in magnetic flux.
- * It is working on the principle of Josephson effect.
- * It consists of superconducting ring with two Josephson junctions in parallel. They are capable of measuring magnetic fluctuations of the order of 10^{-18} T.
- * Super current is branched and passing through the insulating junction X and Y .
Cooper pairs are tunneling through the junction. The super current emerging from the junction are coherent and they interfere. The net super current changes periodically with the changes in magnetic flux. Even a small change in magnetic flux can be detected accurately.



Applications of SQUID

- * SQUIDs are used as very sensitive magnetometer to measure a minute change in magnetic field in the order of 10^{-18} T.
- * used to detect the presence of ships, submarines, by detecting a small disturbances in the earth magnetic field.
- * It is used to measure the weak magnetic pulse generated by heart, brain in their pathological analysis.
- * Principle of SQUID is applied in MRI for the investigation and diagnosis of various diseases.
- * Used to explore the oil deposits and other mineral deposits in different parts of the world.

High Temperature Superconductors: HTC superconductor

- * Substance having T_c around or below 24K are low T_c superconductors.
- * Substance having T_c above 24K are high T_c superconductors.
- * All known high temperature superconductors are type II.

Eg: Yttrium barium Copper oxide (Y-Ba-Cu-O) – $T_c = 93$ K

Mercury thallium barium calcium copper oxide (Hg-Tl-Ba-Ca-Cu-O) – $T_c > 138$ K

Applications:

- * used to produce powerful magnets
- * used to detect infrared radiations
- * used for diagnosis with the help of MRI.

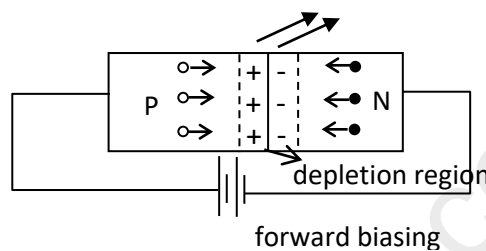
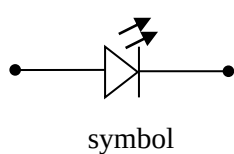
Photonics

Photonics is that branch of Science which deals with the production, control and detection of photons. In Photonics, photons have more or less the same roles as electrons in Electronics. Photonic devices have a number of advantages over electronic devices because of the very high speed of light. So the information transmitted photonically can travel very long distances within a very short time.

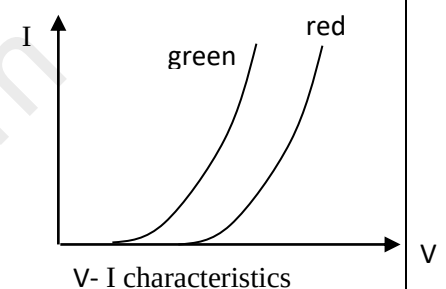
Solid State Lighting (SSL): Solid state lighting deals with devices which gives light output by carrying current. It is a type of lighting that uses mainly LEDs. This type of lighting has higher efficiency, reliability and environmentally friendly technology compared to the conventional incandescent lighting.

Light Emitting Diode (LED)

- * LED is a heavily doped pn junction diode of suitable materials that emit light when it is forward biased.



Block diagram of LED



Working

- * During forward bias, free electrons from conduction band of n region, and holes from valence band of p region moves towards depletion region and recombine together.
- * During recombination, electromagnetic radiation is emitted with energy equal to band gap energy.

$$E_g = h\nu = \frac{hc}{\lambda} \text{ where } \nu \text{ is frequency; } c \text{ is the velocity of light and } \lambda \text{ is the wavelength.}$$

- * The band gap energy determines the colour or wavelength of the emitted light.

Applications

- * used as indicator light.
- * used in remote sensor.
- * used for fancy light and decoration.
- * used in optical communication.

Advantages

- * low power consumption.
- * has lower life.
- * smaller size.

Disadvantages

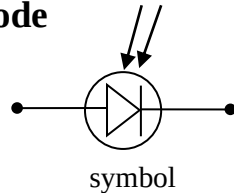
- * Output is low.
- * Intensity of light is small.

- * Incoherent light is produced.

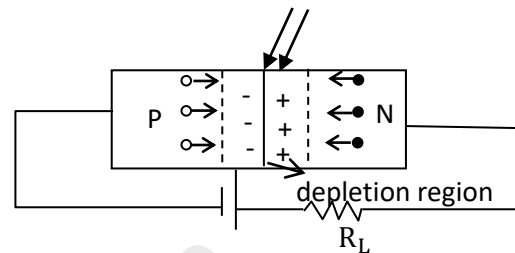
Photo detectors

A photo detector is a device which converts optical signal to electrical signals. These devices are widely used in optical communication systems. They should have high sensitivity, low noise, low cost, low response time, wide bandwidth etc.

Junction Photodiode



- * This is a reverse biased p-n junction semiconductor device which converts light into current.

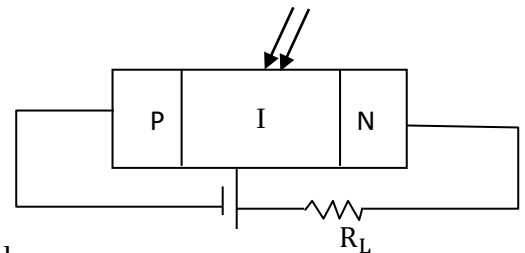


Working:

- * In reverse bias condition, a small current is produced due to the flow of minority carriers from P and N sides. This current is called **dark current**.
- * If light is incident at the junction (depletion region), large number of free electrons and holes are created known as photocarriers.
- * Due to the flow of these free electrons and holes, a current is created called photocurrent.
- * Magnitude of current depends on the intensity of light incident on it.
- * **Uses:** Used in CD players, smoke detectors, remote control devices.

P-I-N Photodiode (P-type – intrinsic – N-type)

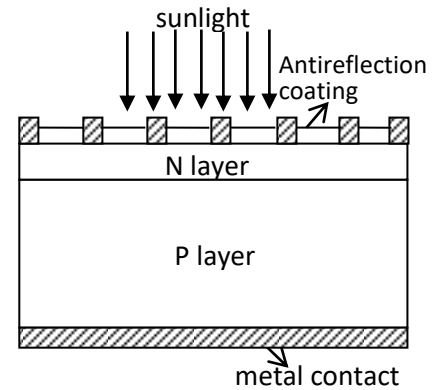
- * In PIN photodiode, a thick intrinsic semiconducting layer is inserted between heavily doped P and N region. This is to improve the sensitivity of the photodiode.
- * When light is incident on the diode, electrons are excited from the valence band to the conduction band producing a large number of electron – hole pairs.
- * Because of its large width, the intrinsic layer absorbs very large number of incoming photons compared to the P and N regions.
- * This increases the photocurrent and improves the efficiency, speed and sensitivity compared to that in photodiode.



Solar cell (Photovoltaic cell)

- * Solar cells are PN junction diode, it absorbs radiant energy or sunlight and converts it into electrical energy.
- * Solar cell consists of a heavily doped p-n junction.

- * It has a large surface area to receive large amount of sunlight.
- * Top N-layer is very thin, to allow solar radiations to reach the p-n junction.



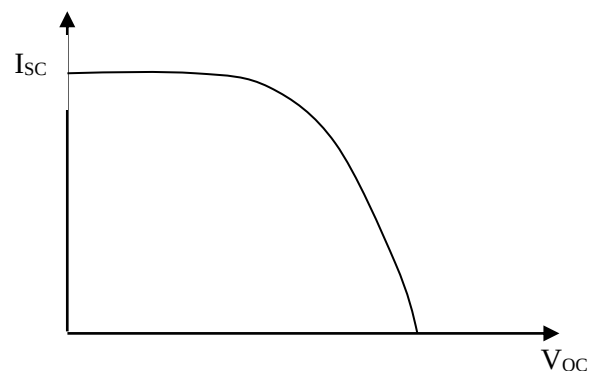
- * Connections are made from bottom P-layer and top N-layer using metal contacts. An antireflection coating is provided on the top layer to prevent loss of light by reflection.

Working

- * Solar cell is a p-n junction diode under zero bias. At zero bias, free electrons flow from n-region to p-region. These free electrons will recombine with holes in the p-region and become bound electrons.
- * When solar radiation are incident at pn junction, more electron-hole pairs are formed producing an electric current in the external circuit. Thus solar energy is converted into electrical energy.
- * Total output voltage can be increased by connecting a number of solar cells in series. A solar panel is an array of a number of solar cells connected together.

VI characteristics of a solar cell

- * At open circuit voltage V_{OC} (external resistance R_L is very high), current is zero.
- * At short circuit current I_{SC} , current is maximum and terminal voltage will be zero.
- * As R_L increases, the current remains constant up to a certain load. Only beyond that load, current decreases. Hence solar cell is called a constant current source.



$$\text{Fillfactor} = \frac{P_{\max}}{V_{oc} \cdot I_{sc}}$$

$$\text{Conversion efficiency} = \frac{P_{\max}}{E \cdot A}$$

Where E-input light irradiance , A –solar cell surface area

Fibre Optics

- * Optical fibre is a cylindrical wave guide system which can be operated at optical frequencies.
- * It consists of three parts.

(i) **Core** is the innermost cylindrical region. Refractive index of the core n_1 is made slightly higher than cladding.

It is made of germanium doped silica glass.

Light wave can propagate through the core.

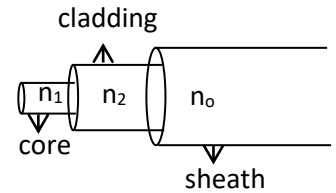
The size of the core is about $10 - 50\mu\text{m}$.

(ii) **Cladding**: A core is surrounded by another cylindrical shell

called cladding. Its refractive index is made smaller compared to the

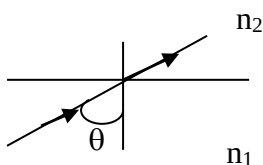
core. It is made of nearly pure silica glass. Purpose of cladding is to keep the light waves within the core itself by total internal reflection. The size of cladding is about $125\mu\text{m}$.

(iii) **Sheath**: Outermost cylindrical layer is called sheath. It is an opaque layer used to provide protection to the fibre. It is made of plastic material. Its size is about $150\mu\text{m}$.

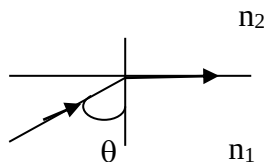


* Working principle of propagation of light through optic fibre – Total internal reflection

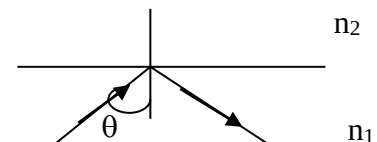
When a ray of light travel from a denser medium of refractive index n_1 to a rarer medium of refractive index n_2 , for a certain value of angle of incidence in the denser medium the refracted ray just grazes the surface of separation. This angle of incidence in a denser medium is called critical angle. If the angle of incidence θ is greater than the critical angle ϕ_c , the ray is totally reflected back into the denser medium. This phenomenon is known as total internal reflection. It is the phenomenon of total internal reflection that is responsible for propagation of light through optical fibre. By multiple total internal reflections, the light is transmitted through the fibre.



$$\theta < \phi_c$$



$$\theta = \phi_c$$



$$\theta > \phi_c$$

Types of optic fibre

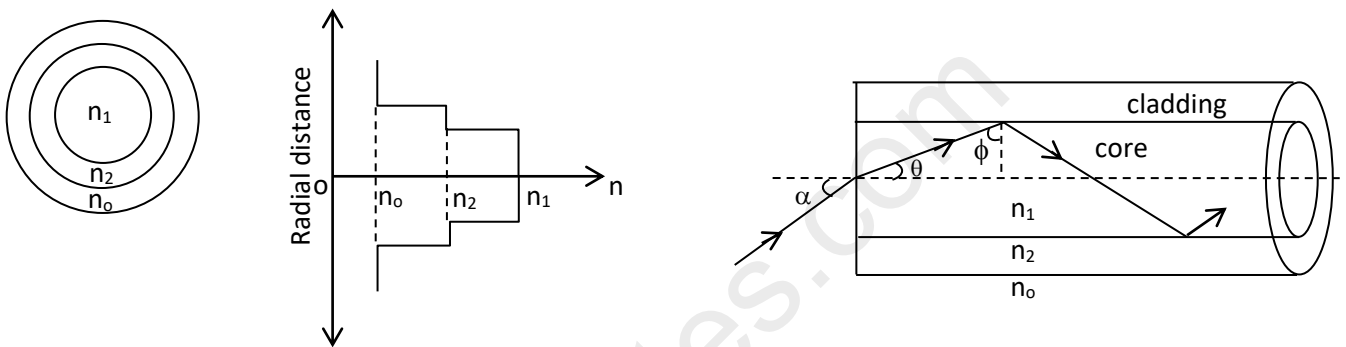
Step - index fibre

- * Refractive index of the core is uniform throughout and an abrupt transition at the core boundary to a region of lower refractive index.

- * Light travels in a Zig- zag path.

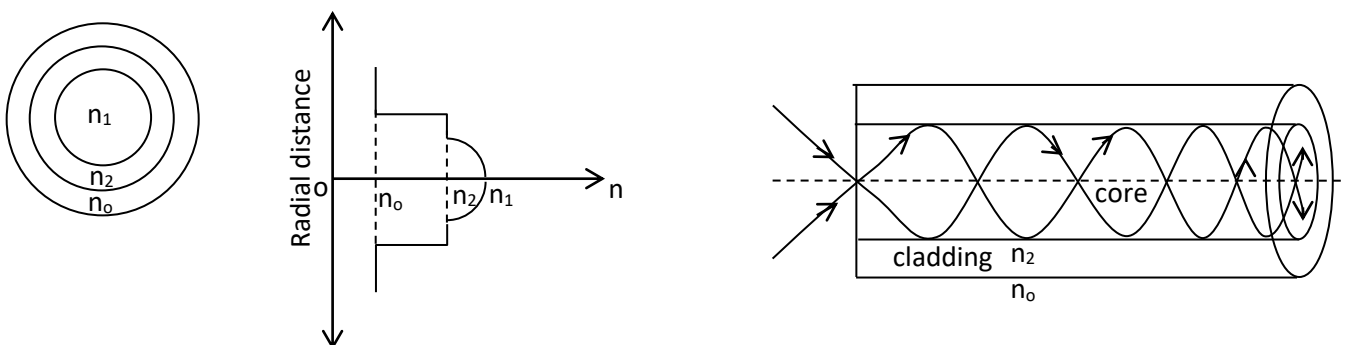
*only one signal is passed through a single mode step index fibre and its core diameter is small.
 V_{number} less than 2.405

- More than one signal is passed through multi mode step index fibre and its core diameter is large.
 V_{number} greater than 2.405
- No of modes for multimode step index fibre = $(V_{\text{num}}^2) / 2$
- * Shape of index profile (graph with refractive index in x-axis and distance from the axis of fibre in y-axis) is like a step. Hence it got the name step-index fibre.



Graded index fibre

- * This type of optical fibre consists of core with varying refractive index and cladding of constant refractive index and $n_1 > n_2$.
- * n_1 is maximum at axis of the core and decreases radially outwards.
- * Light entering through one face of fibre is travelling in a parabolic path by undergoing a number of gradual total internal reflection.
- No of modes = $(V_{\text{num}}^2) / 4$



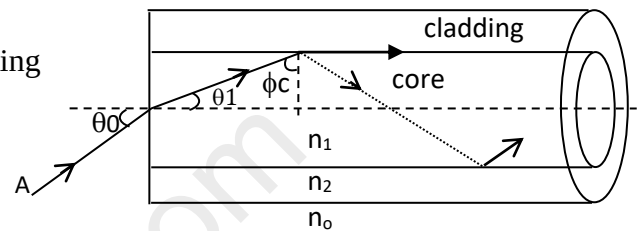
Numerical Aperture of an optical fibre

- * It is the measure of its light gathering capacity of an optical fibre.
- * It is the sine of the maximum value of acceptance angle.

* Numerical aperture $NA = \sin \alpha_m = \sqrt{n_1^2 - n_2^2}$

Consider a step index fibre made of core with refractive index n_1 and a cladding of refractive index n_2 . Here $n_1 > n_2$.

- * Let a light ray is incident at an acceptance angle θ_0 .
- * It is refracted along BC at θ_1 .
- * After refraction, the ray is incident at the core-cladding interface at critical angle ϕ_c .



by Snell's law

$$\frac{\sin \theta_0}{\sin \theta_1} = \frac{n_1}{n_0}$$

$$n_0 \sin \theta_0 = n_1 \sin \theta_1 \rightarrow (1)$$

From figure, $\theta_1 = \frac{\pi}{2} - \phi$ ----- (a)

So eqn (1) becomes, $\sin \theta_0 = \frac{n_1}{n_0} \sin \left(\frac{\pi}{2} - \phi \right)$

$$\sin \theta_0 = \frac{n_1}{n_0} \cos \phi \rightarrow (2)$$

$$\sin \left(\frac{\pi}{2} - \phi \right) = \cos \phi$$

The minimum value of ϕ for total internal reflection $\phi_{\min} = \phi_c$

When ϕ is minimum θ_1 is maximum from eqn (a) so θ_0 is also maximum

$$\therefore \sin \theta_{0\max} = \frac{n_1}{n_0} \cos \phi_c \rightarrow (3)$$

From fig at the core cladding interface, from snells law we have

$$n_1 \sin \phi_c = n_2 \sin 90$$

$$\sin \phi_c = \frac{n_2}{n_1}$$

We know that, $\cos \phi_c = \sqrt{1 - \sin^2 \phi_c} = \sqrt{1 - \frac{n_2^2}{n_1^2}}$

$$\text{i.e., } \cos \phi_c = \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

$$\text{or } \cos \phi_c = \frac{\sqrt{n_1^2 - n_2^2}}{n_1} \rightarrow (4)$$

Substituting, (4) in (3), $\sin \theta_{0max} = \frac{n_1}{n_0} \frac{\sqrt{n_1^2 - n_2^2}}{n_1}$

$$\text{Or NA} = \sin \theta_{0max} = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

The refractive index of air is unity, $n_0 = 1$, then $\text{NA} = \sin \theta_{0max} = \sqrt{n_1^2 - n_2^2}$

Acceptance angle(θ_{0max})

Maximum angle that a light ray can have relative to the axis of the fibre and propagate down the fibre.

Acceptance cone

The light ray contained within the cone having full angle $2\theta_{0max}$ are accepted and transmitted. This cone is called the acceptance cone.

Relative refractive index of a fibre or fractional refractive index (Δ)

It is the ratio of the difference between refractive indices of the core and cladding of a fibre to the refractive index of the core. $\Delta = \frac{n_1 - n_2}{n_1}$

Relation between NA and relative refractive index of a fibre (Δ)

$$\text{NA} = \sin \theta_{0max} = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 - n_2)(n_1 + n_2)}$$

$$\Delta = \frac{n_1 - n_2}{n_1} \quad ; \quad n_1 - n_2 = \Delta n_1$$

$$\text{Since } n_1 \approx n_2, \quad n_1 + n_2 = 2n_1$$

$$\text{Then NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 - n_2)(n_1 + n_2)} = \sqrt{2n_1 \Delta n_1}$$

$$\text{NA} = n_1 \sqrt{2\Delta}$$

V-number(cut off parameter, normalized frequency parameter)

It determines the number of modes a given fibre can support given as

$$V_{\text{num}} = \frac{\pi d}{\lambda} * \text{NA}$$

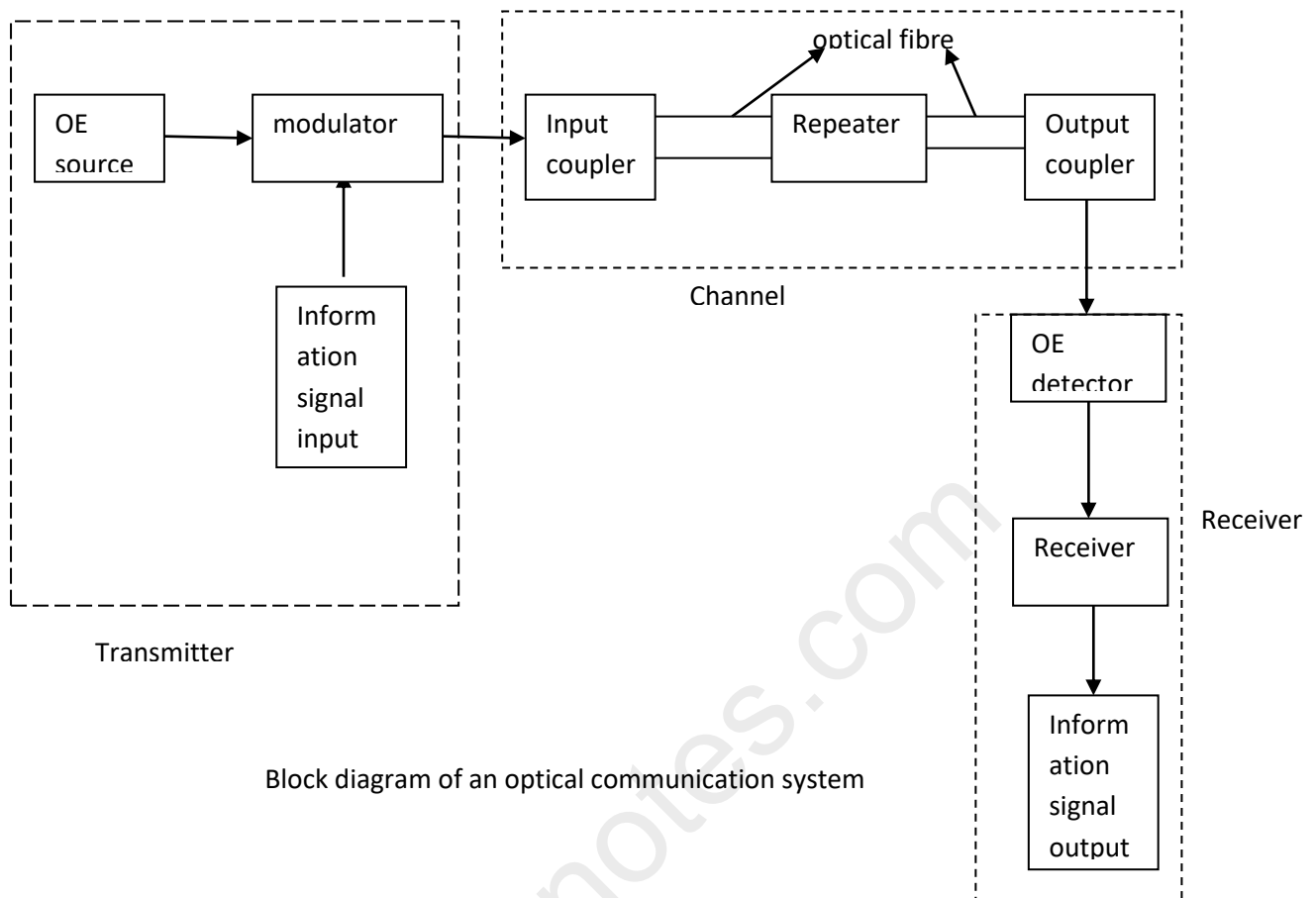
$$V_{\text{num}} = \frac{\pi d}{\lambda} \sqrt{n_1^2 - n_2^2}$$

Advantages of optical fibre

1. Low transmission loss and cost effective.
2. Lesser weight and volume.
3. Immune to electromagnetic interference.
4. Can be used in explosive and high voltage areas.
5. Long lifespan, made of silica and glass so very cheap.
6. Survive corrosive environment and not affected by temperature variations

Fibre optic Communication System (FOS)

It consists of three sections (i) transmitter (ii) receiver (iii) channel (fibre optic cable)



Block diagram of an optical communication system

Information input

This is a transducer which converts the non electrical inputs into actual electrical inputs.

Eg:-video camera, microphone etc.

Transmitter

This stage converts the electrical signal into proper form. The signal is impressed on a carrier wave generated by the optoelectronic source.

Input and Output coupler

Coupler is a device which connects the transmitter output to the channel. At the end also coupling is required to collect the signal from optical fibre and direct it on to the photo detector.

Fibre optic channel

It is the path between the transmitter and the receiver. In FOS the optical signal travels along the cable consisting of single fibre or a bundle of fibres.

Repeater

As optical signal propagates along the length of the fibre, they get attenuated. Repeaters are used to strengthen, restart and retransmitting the signal. A generator or amplifier is used for this purpose.

Optoelectronic detector

The reconversion of an optical signal into an electrical signal takes place at the OE detector. Semi conductor PIN diode or avalanche photodiode are used for this purpose.

Receiver

In this the output photocurrent is filtered to remove any undesired frequency accompanying the signal. After filtering the signal is amplified.

Information output

Finally the electrical signal is converted into proper form by suitable transducers.

Applications of optical fibre**Industrial uses:**

- * Optical fibres are used as sensors to measure or monitor displacement, pressure, temperature, flow rate, chemical composition etc.
- * Used in security alarm system, industrial automation etc.
- * Used to know the level of atmospheric pollution.

Technological uses:

- * Used in cable TV, CCTV, LAN, WAN etc.
- * Internet connections made through optical fibres.
- * Used for transmission of digital data generated by computers.

Medical uses:

- * Optical fibres used in endoscopy.
- * Used to test the tissues and blood vessels which are far below the skin.
- * They are used to measure and monitor blood flow, oxygen saturation level etc.
- * Used for bloodless surgery.
- * Used in angioplasty.

Optical fibre Sensors:

- * It is a device used to measure displacement, force, pressure, temperature etc by detecting the variation of light through fibre with respect to external parameters.
- * There are different types of sensors.

Intensity Modulated Sensor

- * Here a change in physical parameter produces a change in intensity of light through the optic fibre.
- * By measuring this change in intensity of light, the change in physical parameter can be found out.
- * Eg: Force or pressure sensor.

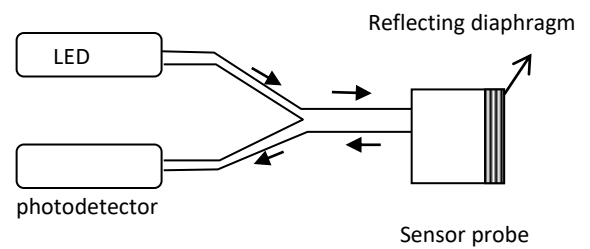
A typical pressure sensor is as shown in figure.

The light from the LED gets reflected from the diaphragm and it is detected by the photodetector.

Changes in external pressure causes the diaphragm to bend leading to changes in numerical aperture of the fibre.

This produces the modulation in the intensity of light transmitted by the fibre. Pressure changes upto 6MPa can be measured accurately.

These types of fibre optic sensors are useful for monitoring pressure changes in arteries, bladder urethra etc. They are also useful for pressure monitoring of gaseous reactants and products in chemical industries.



Phase Modulated Sensor

- * Here a change in physical parameter produces a change in phase of light through the optic fibre which can be detected and calculated for measuring the physical quantity.
- * Eg: Temperature sensor.

Light from a highly monochromatic laser source is split equally by a coupler and sent through the sensing fibre arm and reference fibre arm.

The output light from the two fibres are recombined

at the second coupler. The sensing arm is in direct contact with

the measurand while the reference arm is protected from external parameters. The measurand acts on the sensing arm and causes a change in its length, its refractive index or both. This produces a change in phase of light wave passing through the sensing arm. The light wave through the reference arm is not affected by the measurand. So there appears a phase difference between the two light waves reaching the second coupler. The intensity of light waves emerging as the output depends on this phase difference.

The changes in phase can be detected accurately by interferometer.

